

Simscale Simulations

Arch 573 – Tall buildings Technologies II

Wind Comfort acc. to Bangkok context :

Bangkok is a **tropical, humid city** with generally low-to-moderate average wind speeds. Pedestrian comfort in such climates is often more sensitive to **high-speed gusts** (especially during the rainy season or storms) but also benefits from **moderate breezes** for thermal comfort due to the warm temperatures year-round.

As per BMA – Bangkok Metropolitan Administration:

Comfort Category (Wind)	Comfort Category (Pressure)	Velocity (m/s)	Suitable Activities	Bangkok Applicability
0–2 m/s	0.6-2.4 Pa	Too calm	Sitting/standing only briefly	May feel too warm
2–4 m/s	2.4-9.6 Pa	Comfortable	Sitting/standing long-term	Ideal for shaded areas
4–6 m/s	9.6-21.6 Pa	Pleasant breeze	Walking comfortably	Ideal for circulation paths
6–8 m/s	21.6-38.4 Pa	Windy	Walking okay, but less ideal	Tolerable, avoid near entrances
8–10+ m/s	38.4-60 Pa	Uncomfortable	Risk of discomfort or danger	Should be mitigated

Critical factors taken into consideration:

Wind velocity

3 m/s
7 MPH

Ambient Temperature

25°C
77°F

Surface Temperature

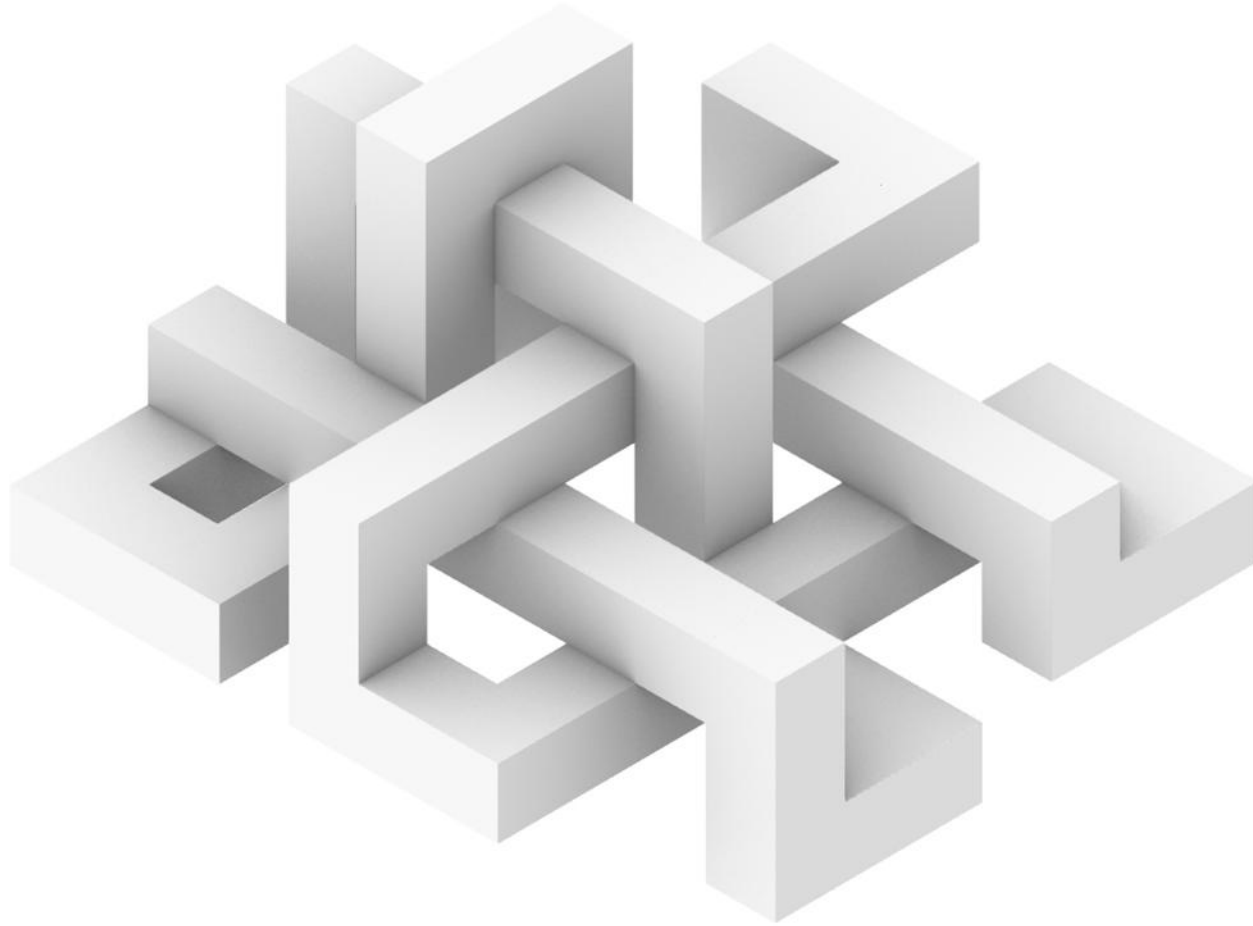
20°C
68°F

Additional Heat source

Human radiated heat
58.2 W/m²

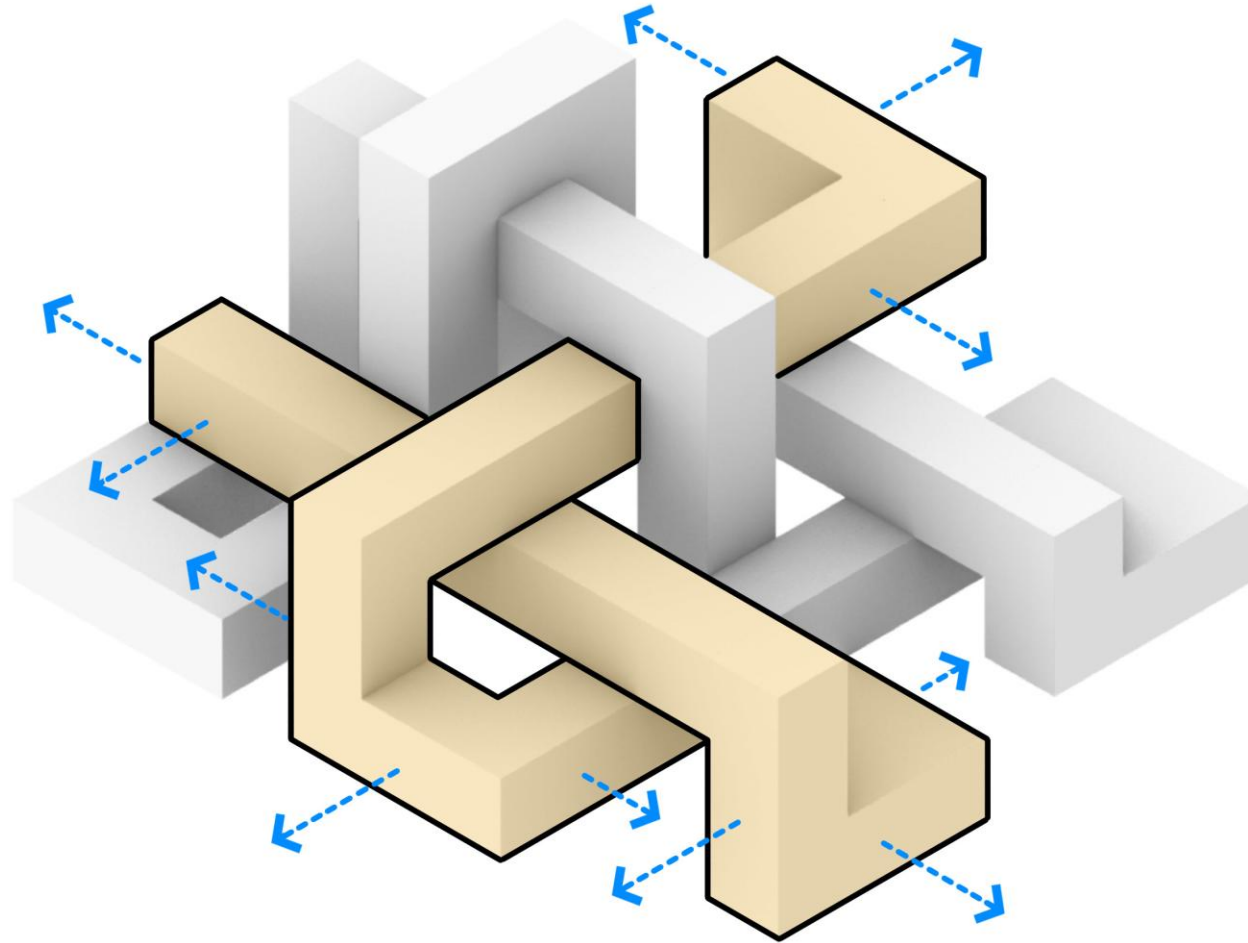
*The weather data is considered as per the month of June with highs of 35°C and lows of 20°C, with relative humidity 81% and winds average of 7MPH

Design Module :



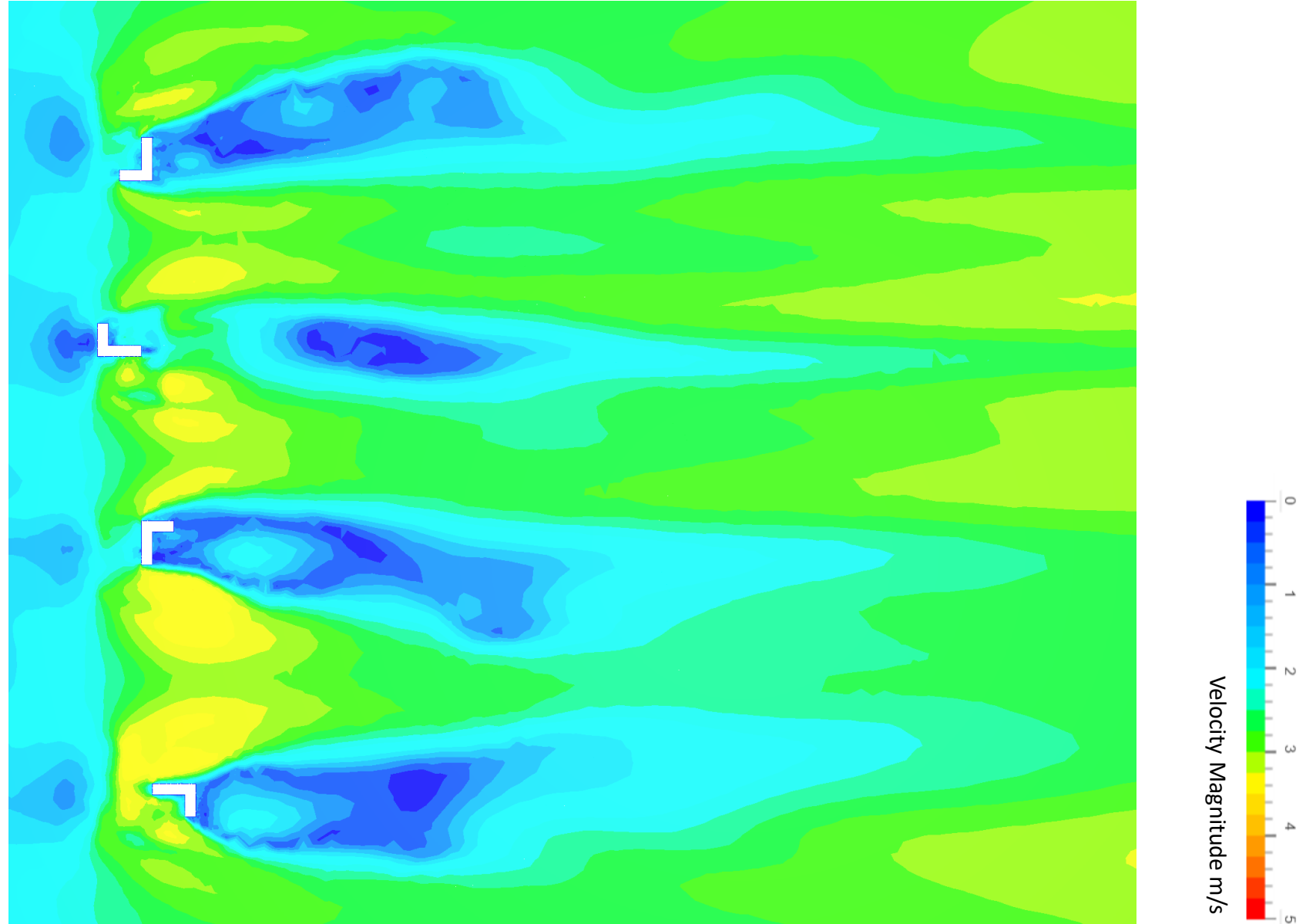
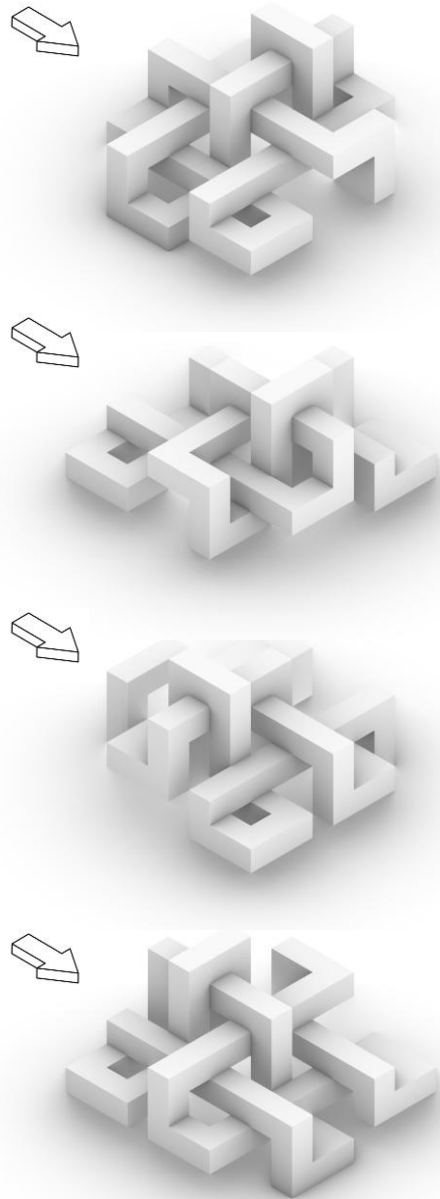
The Design Module inspired by the endless knot – continuous volume flow

Design Module Advantages :

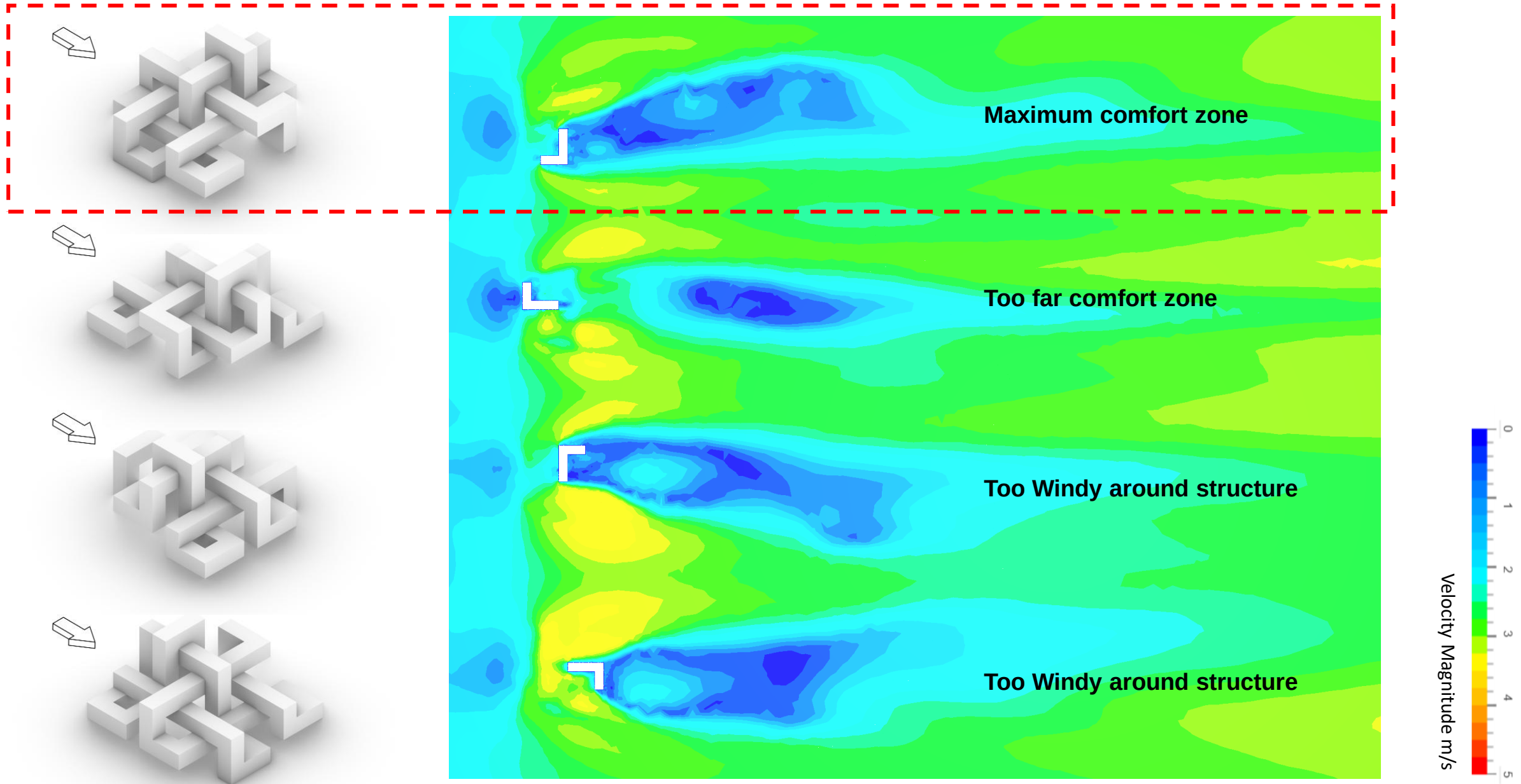


In Tropical Climate, every space has cross ventilation & natural light from all direction

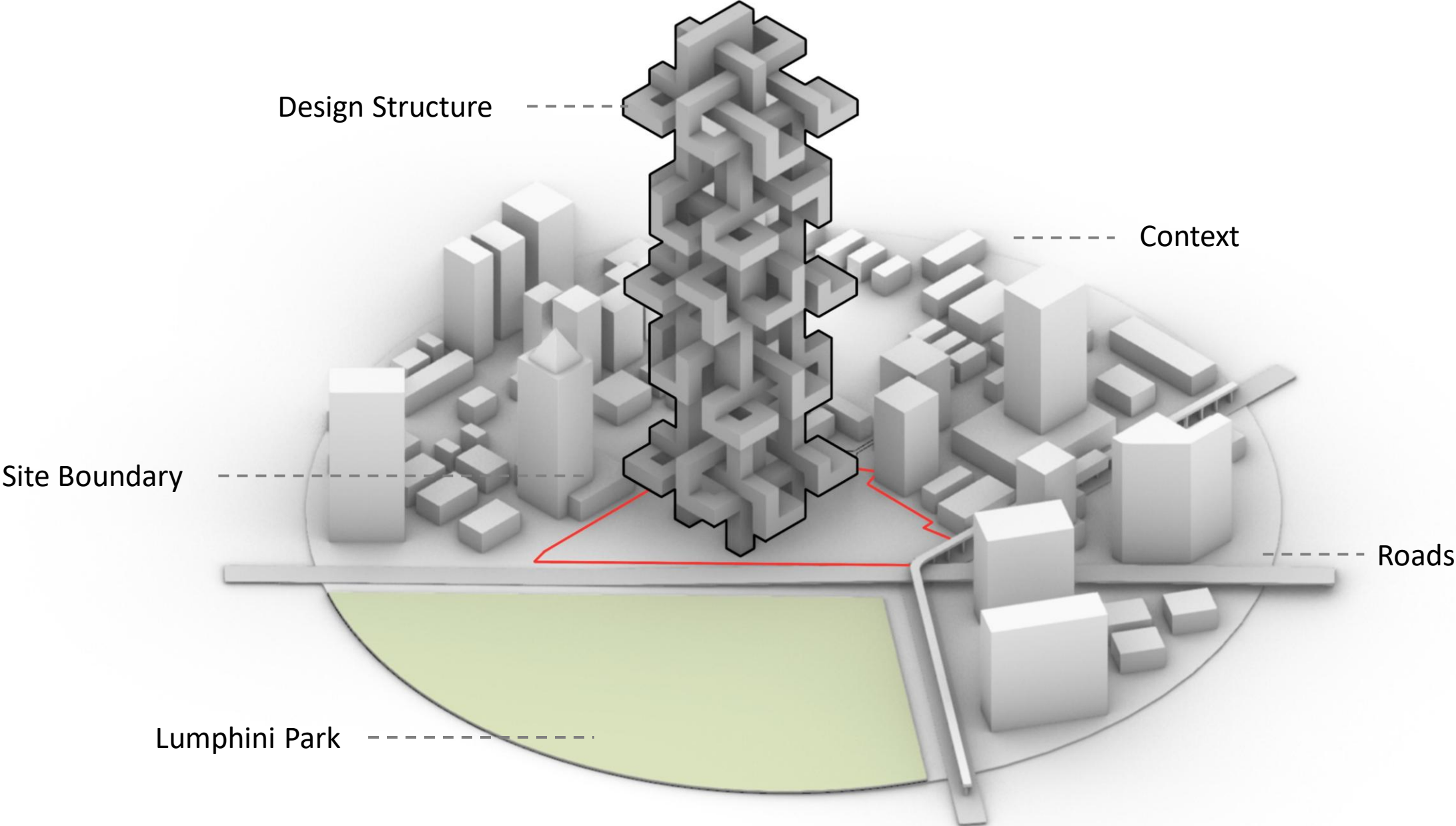
Optimal Orientation for pedestrian comfort at 1.5 M height:



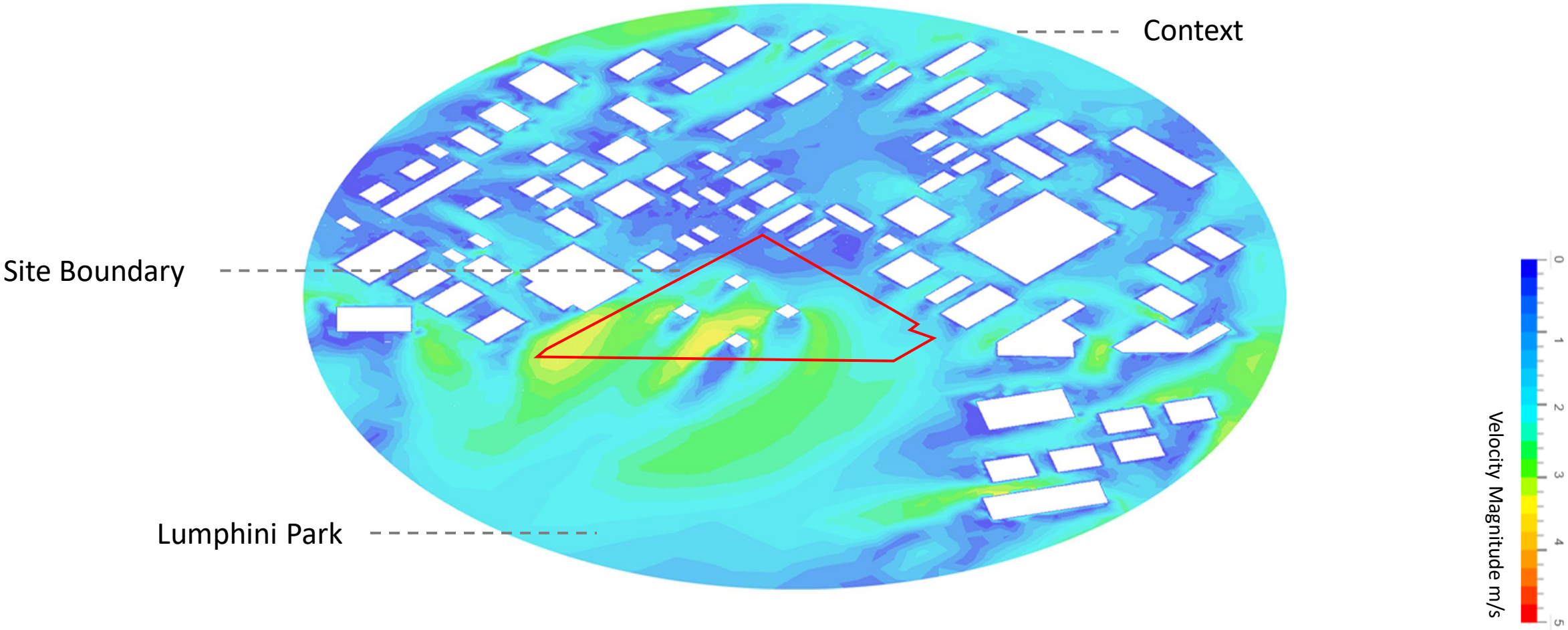
Interpretation for pedestrian comfort at 1.5 M height :



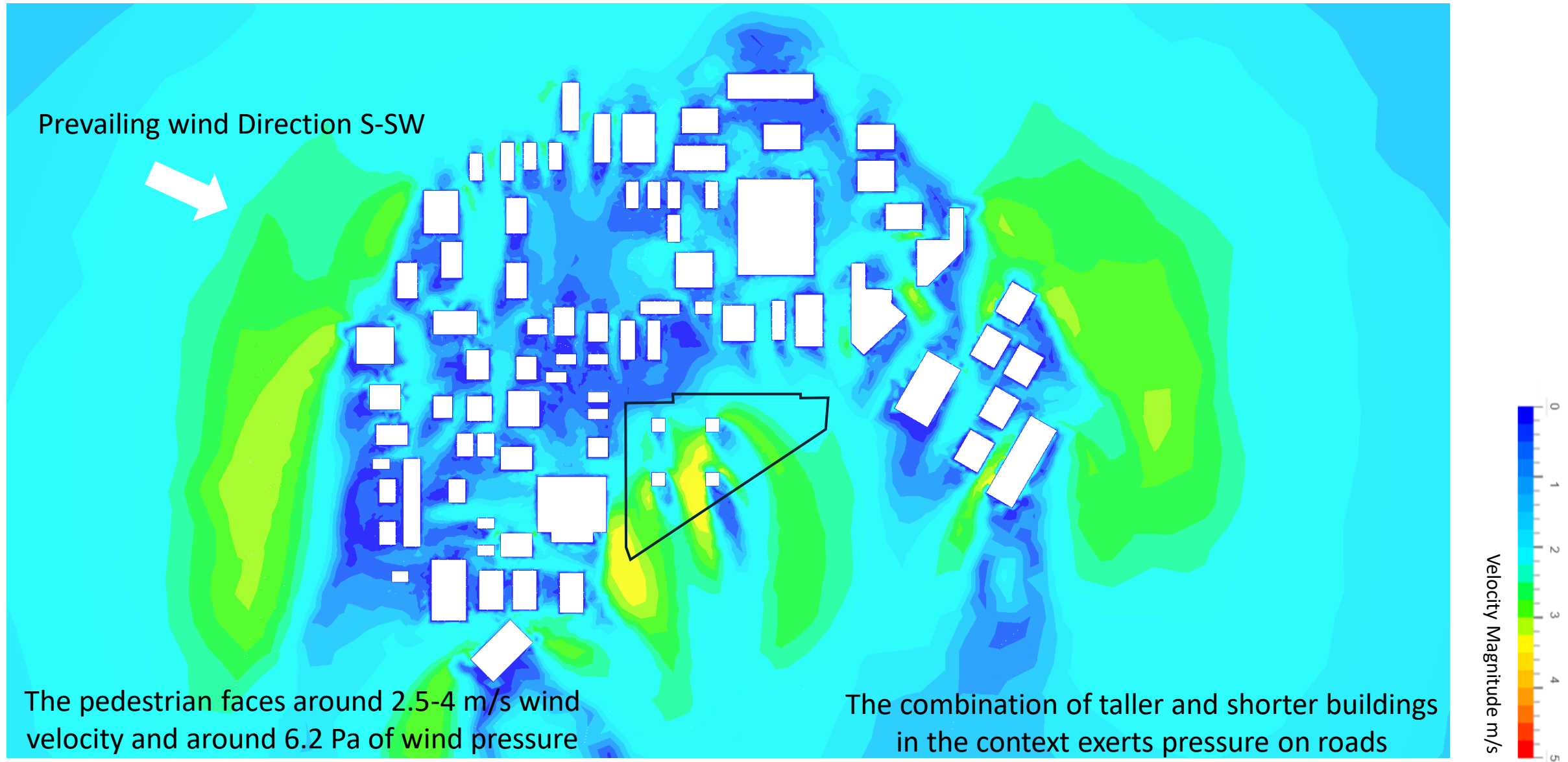
Design Structure with site context :



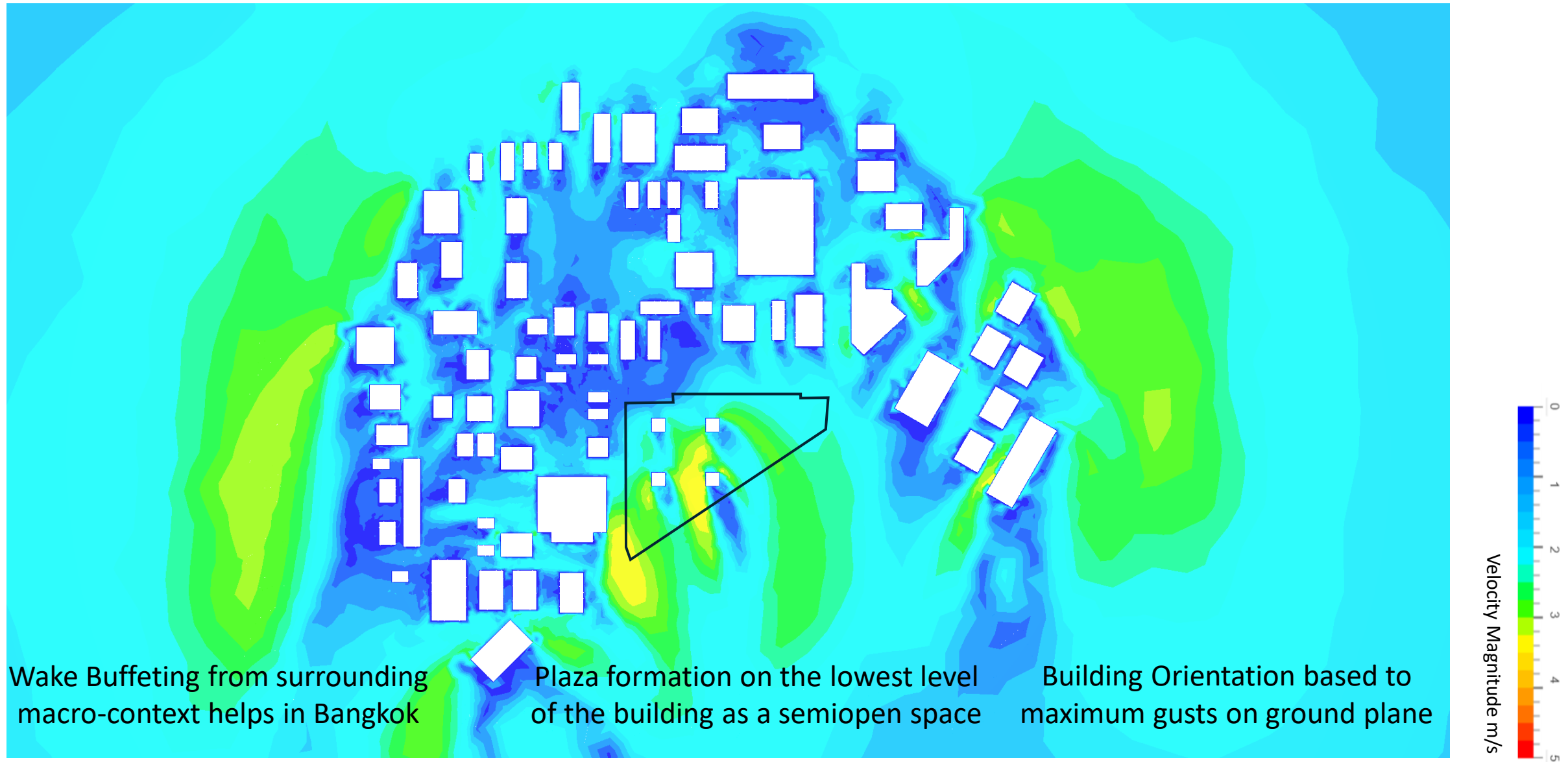
Air movements on site context :



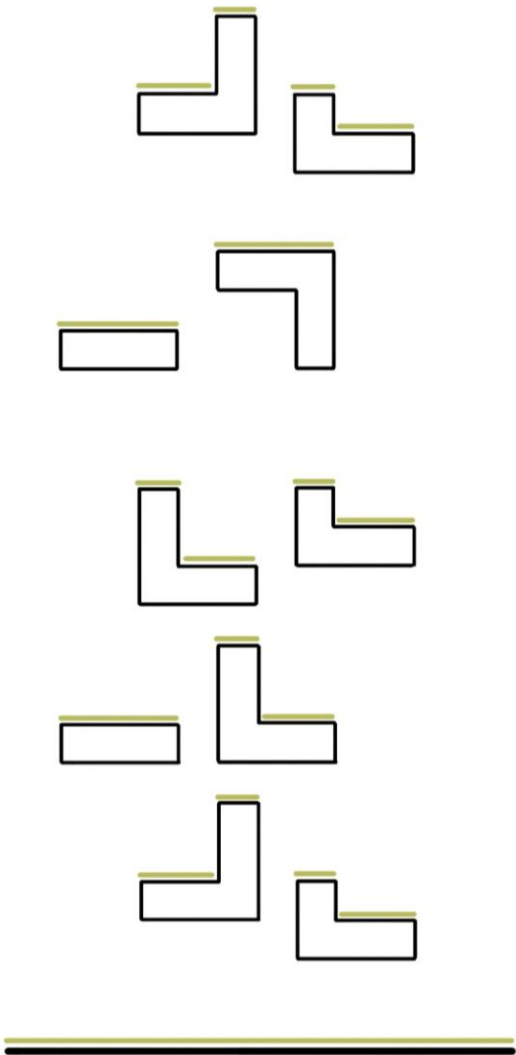
Result Interpretations on the Ground Plane at 1.5 M height :



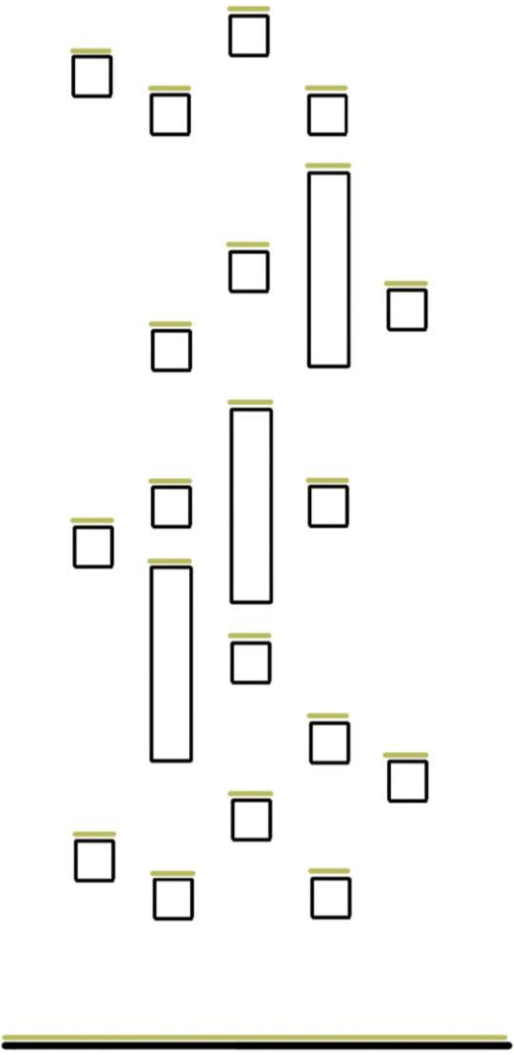
Design Decisions on the Ground Plane :



Sectional Terraces :

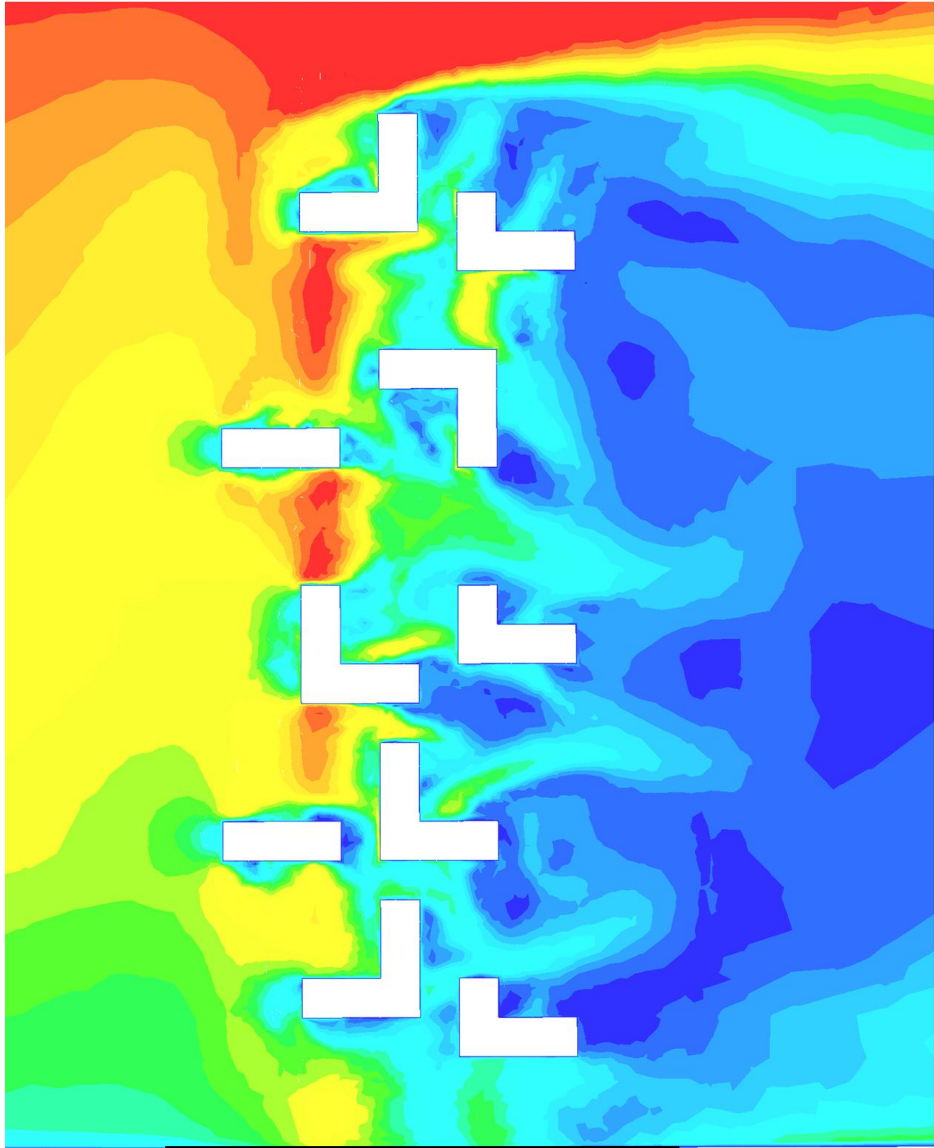


Section 1 with terraces

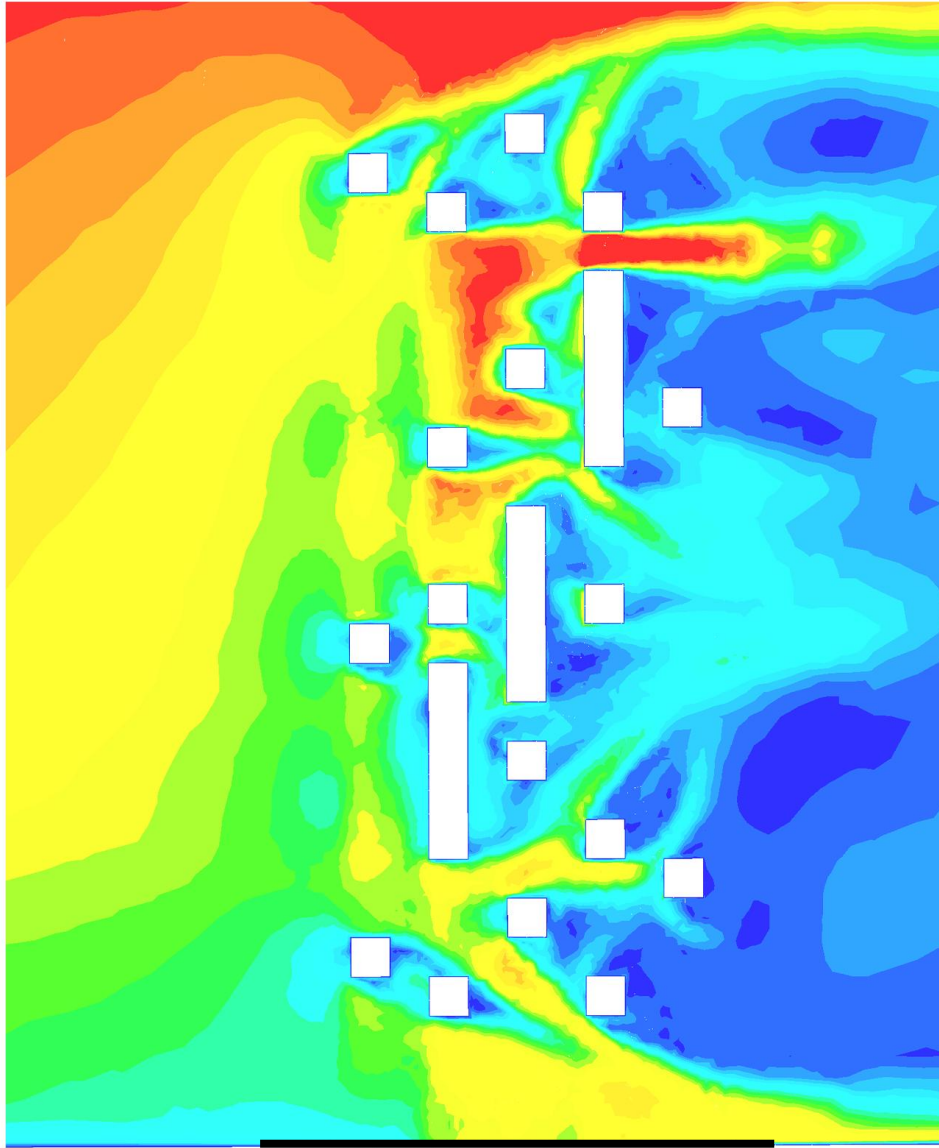


Section 2 with terraces

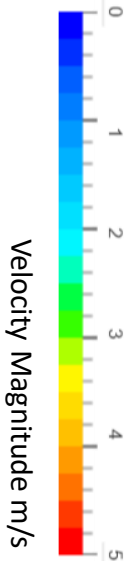
Sectional Terraces External Airflow Results :



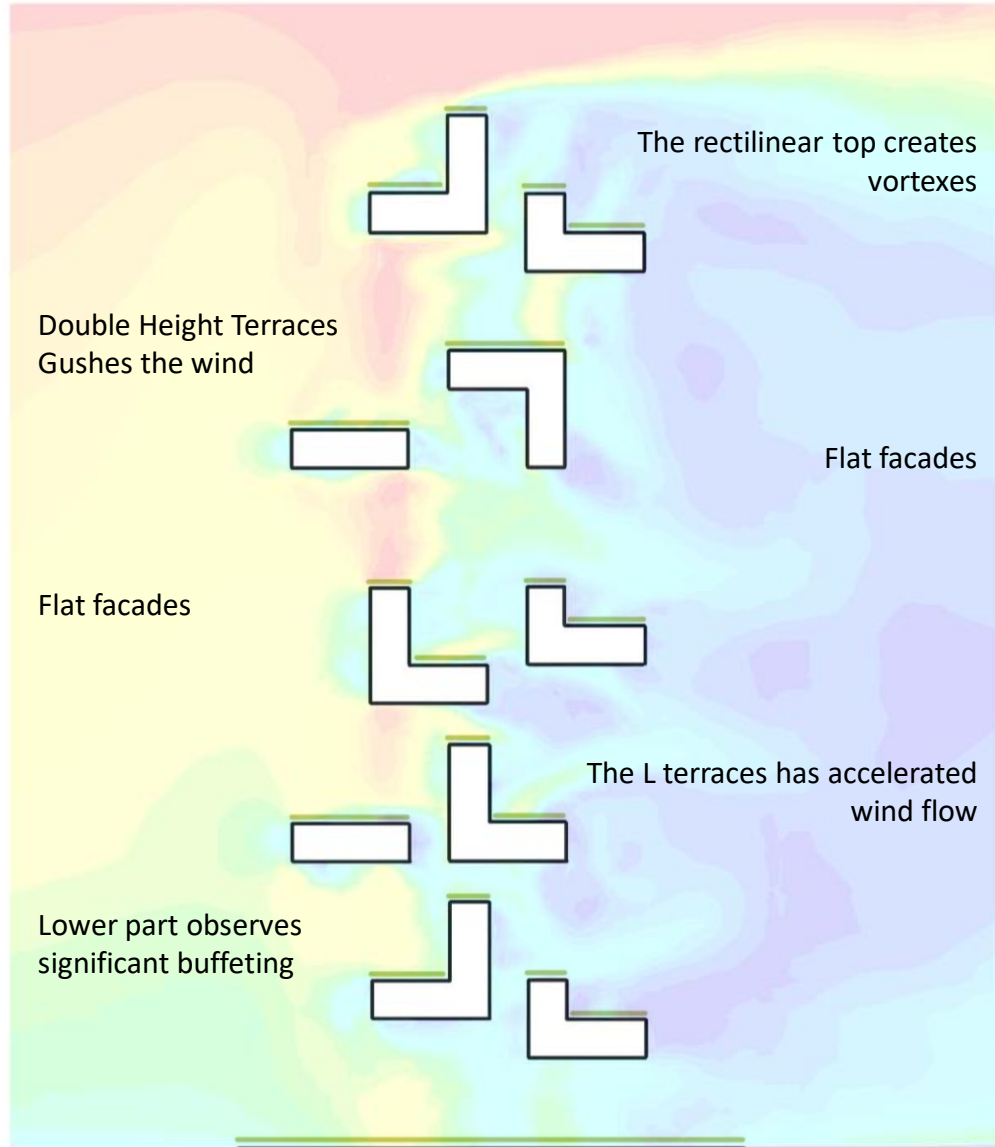
Section 1 – Air movement



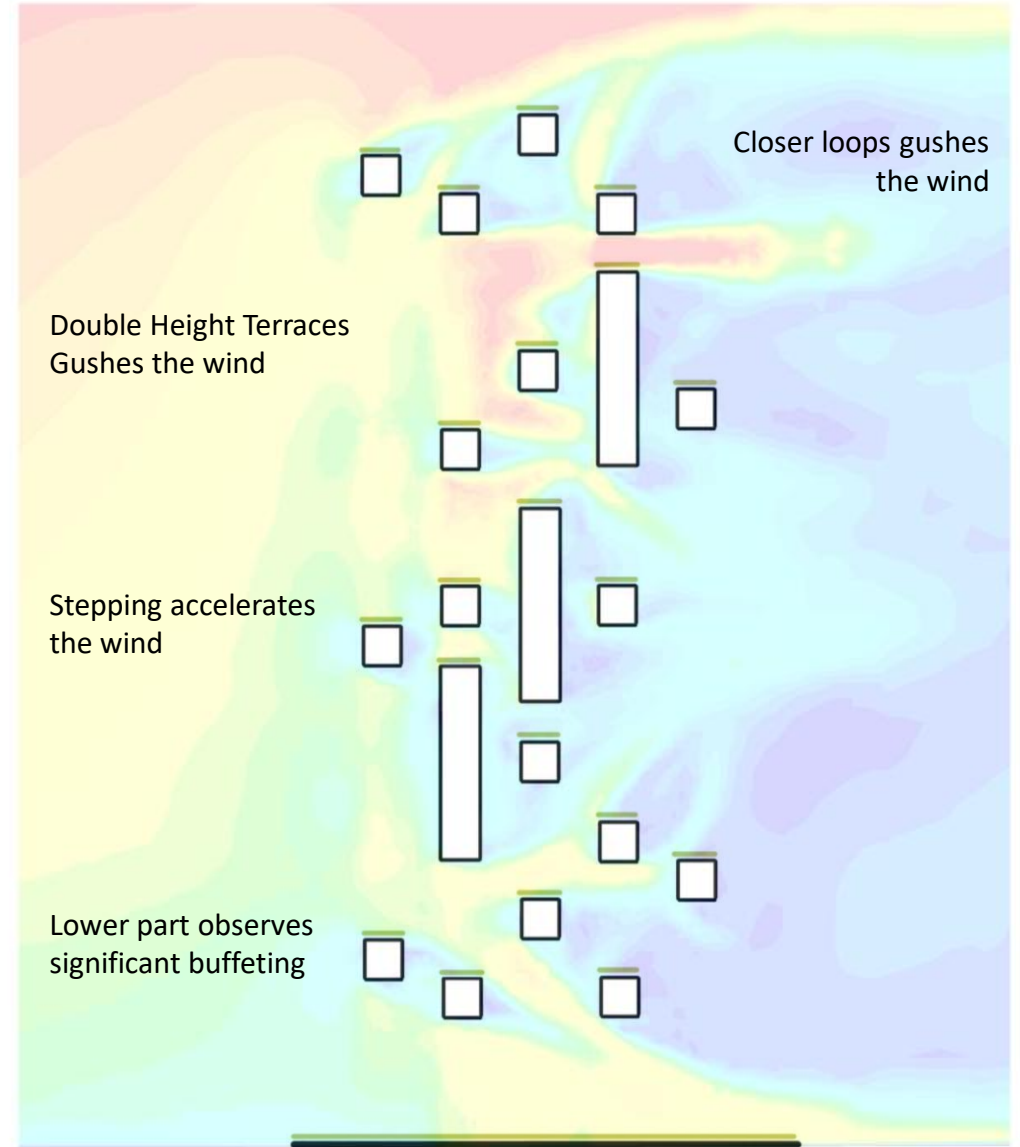
Section 2 – Air movement



Sectional Terraces Result Interpretations :

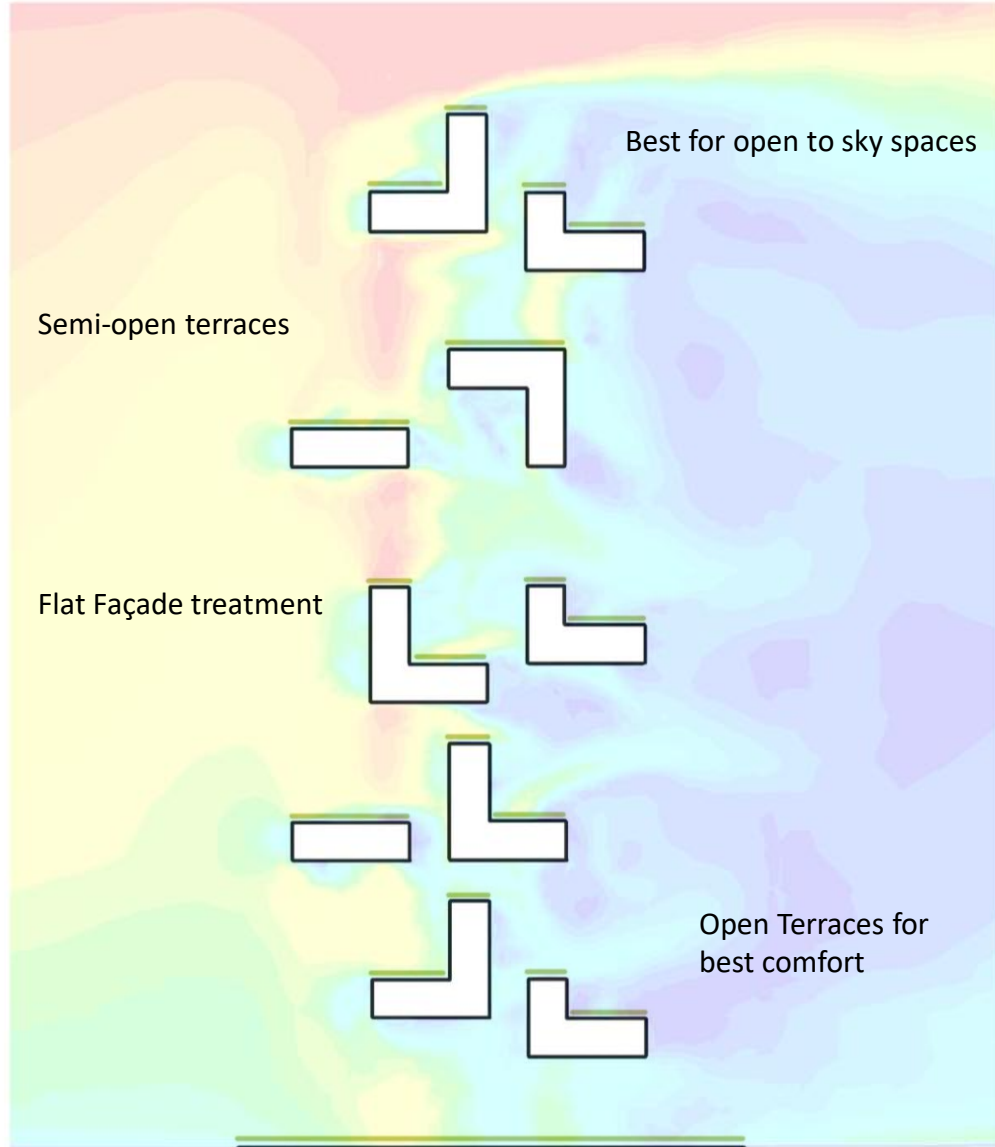


Section 1 – Air movement

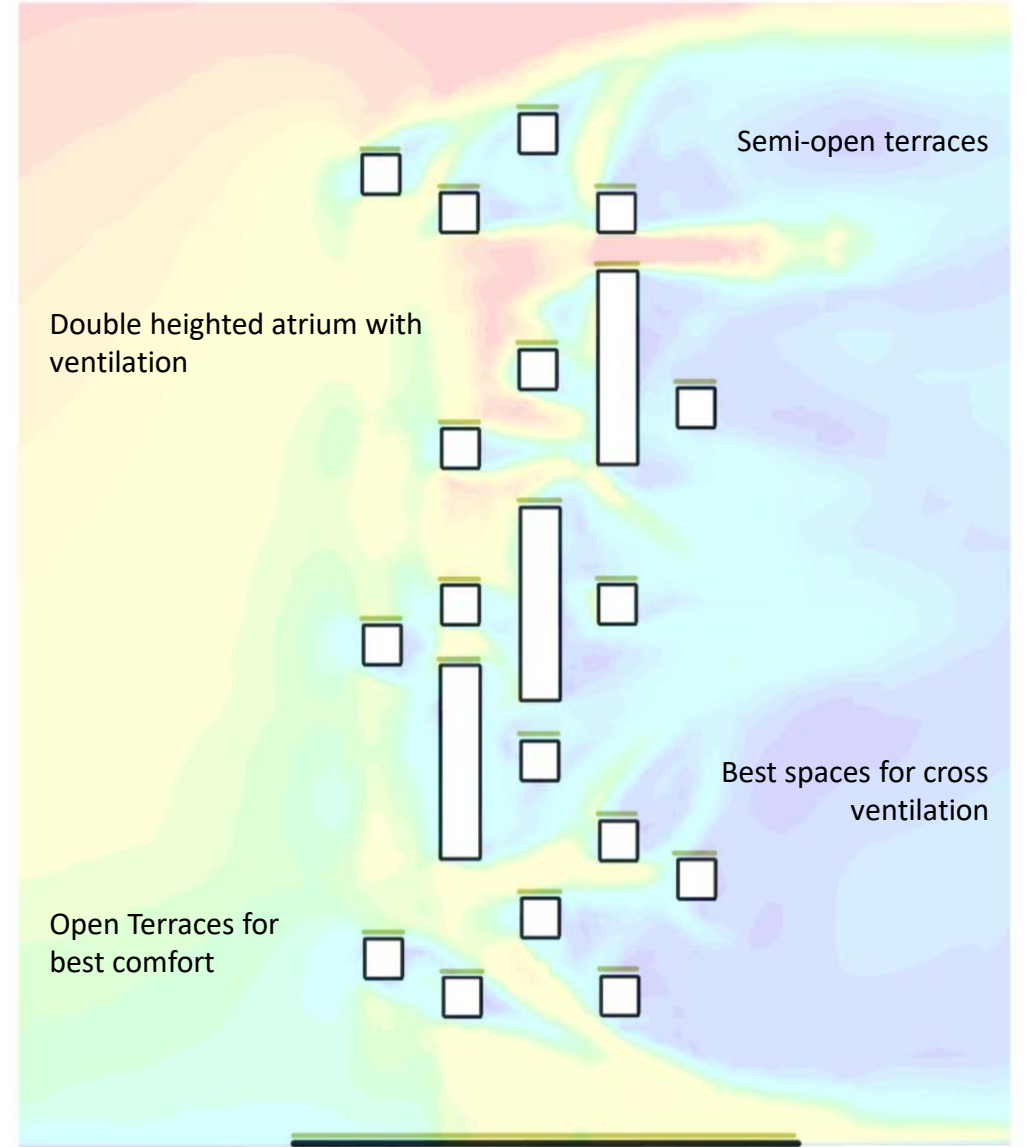


Section 2 – Air movement

Sectional Terraces Design Decisions :

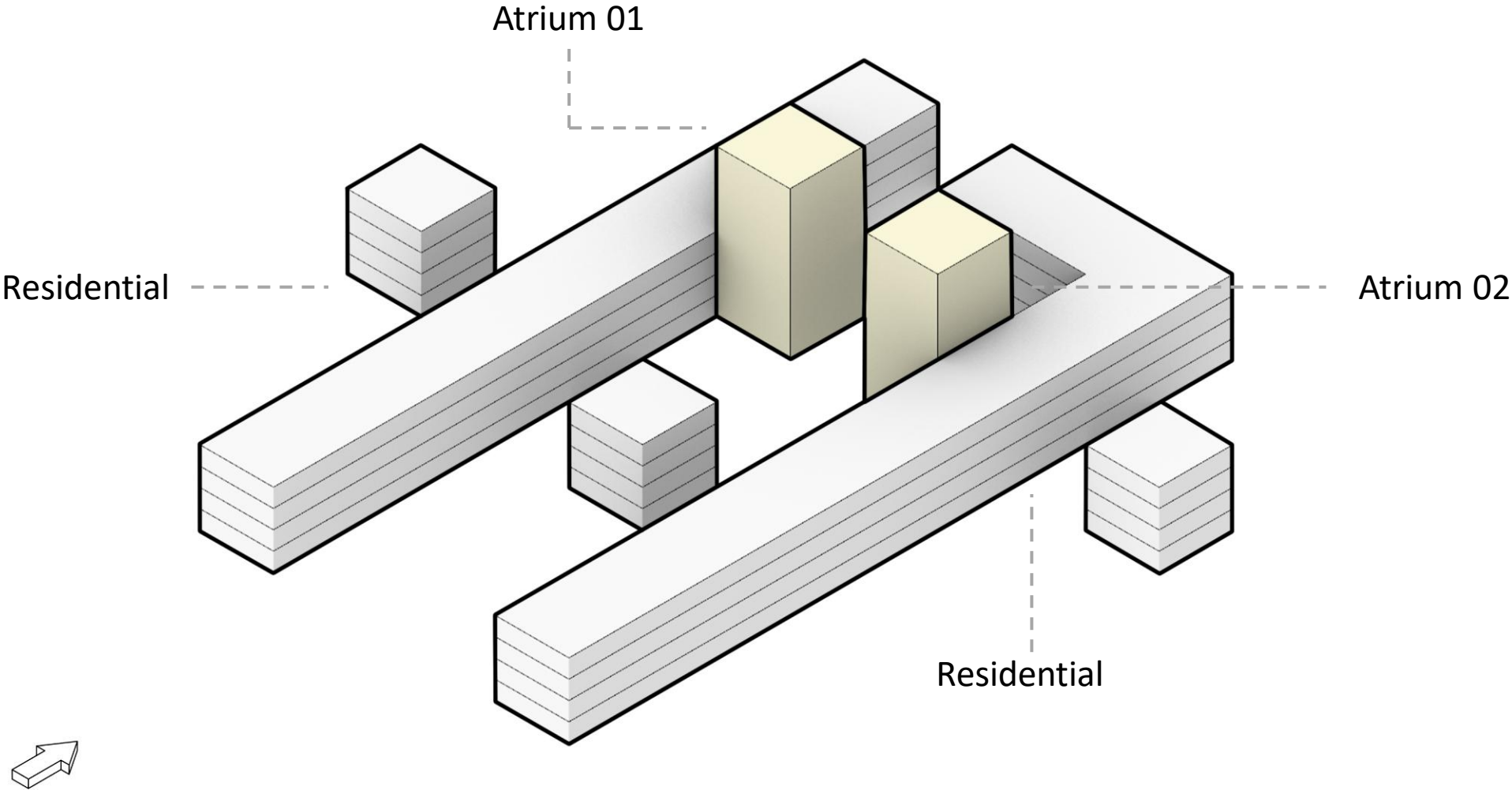


Section 1 – Air movement



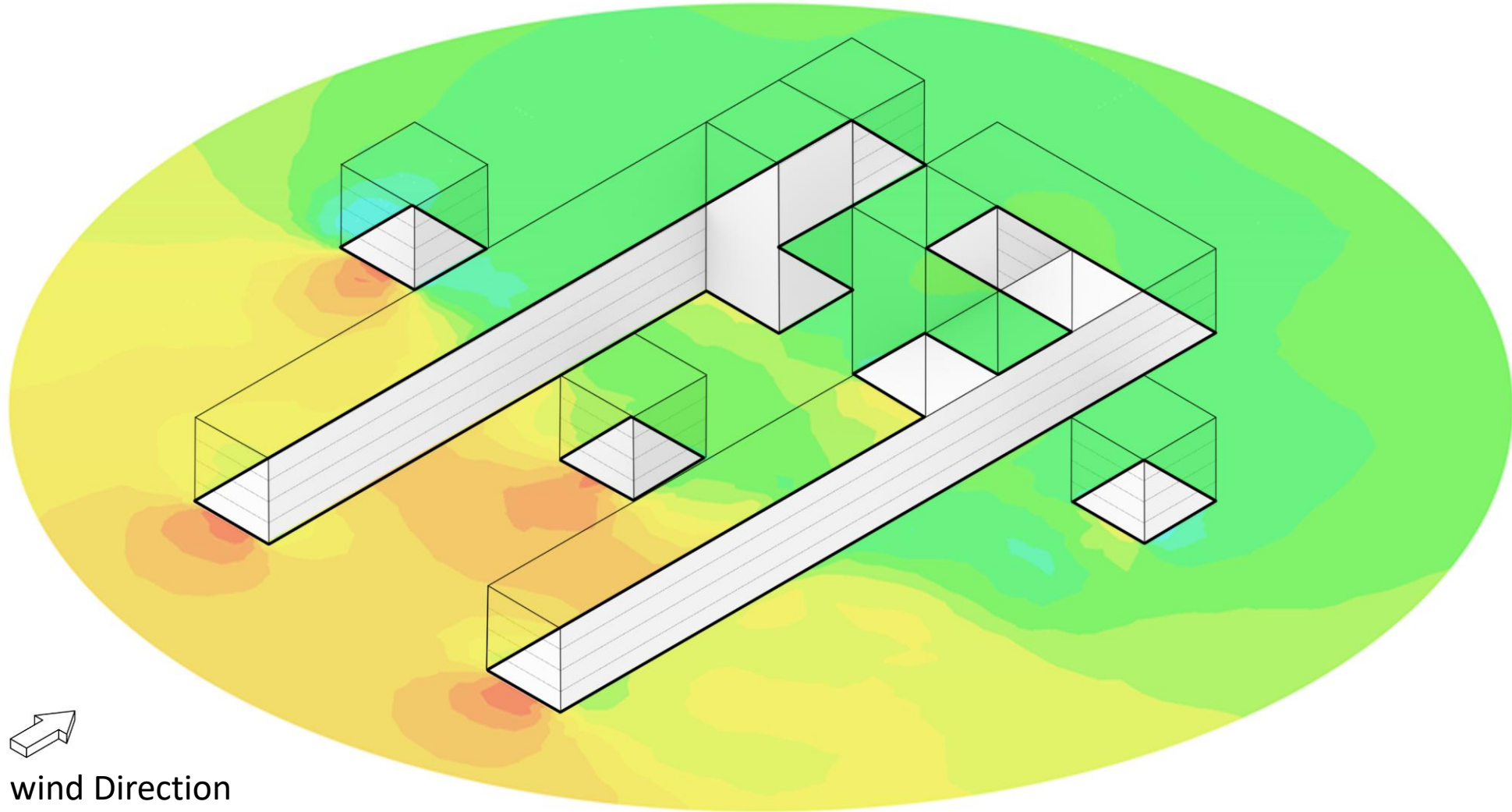
Section 2 – Air movement

Module for Internal Simulations (Level 55 – 165m) :



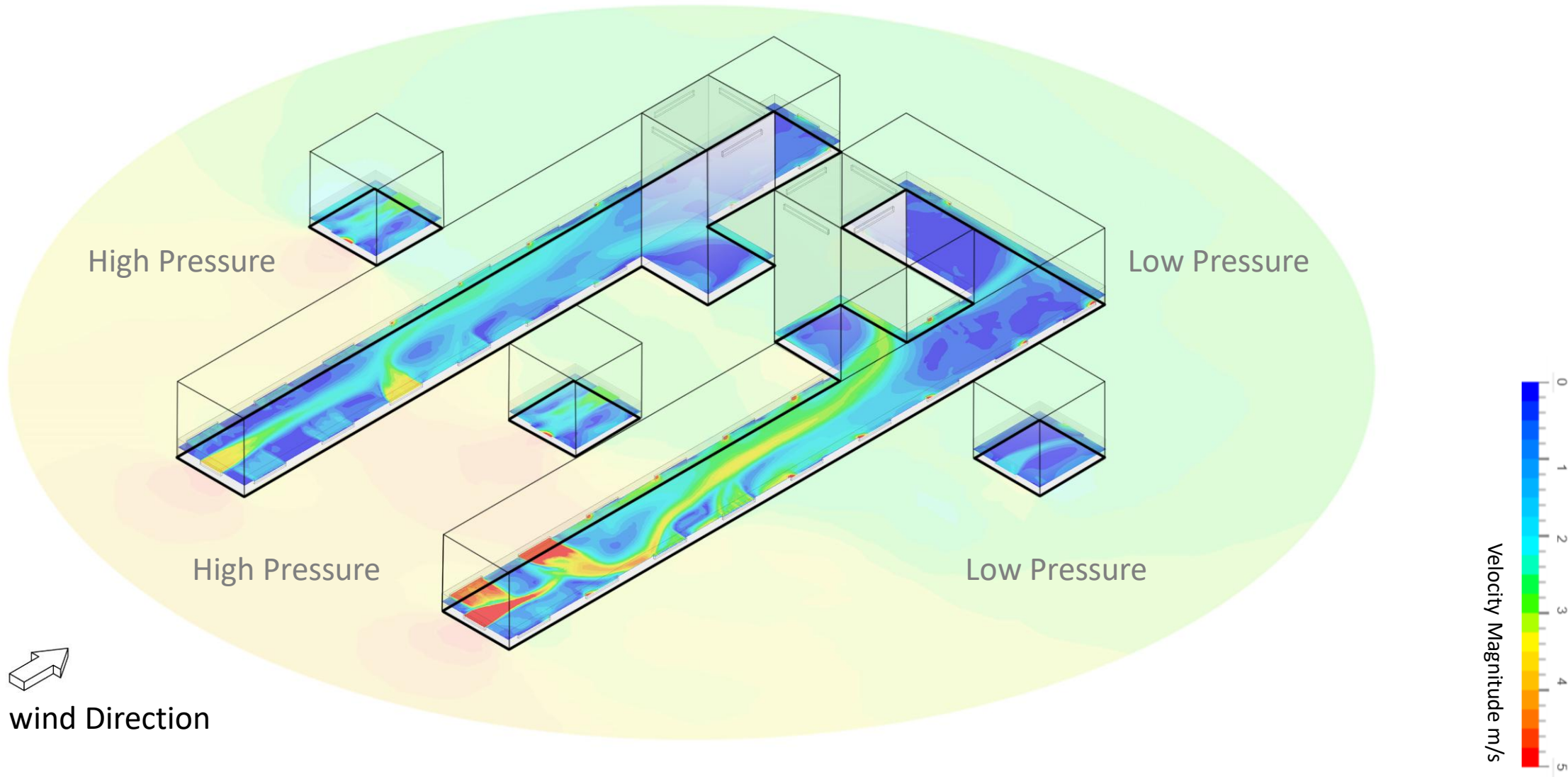
Prevailing wind Direction

External Pressure of the module :



Prevailing wind Direction

Internal Air Flow of the module :



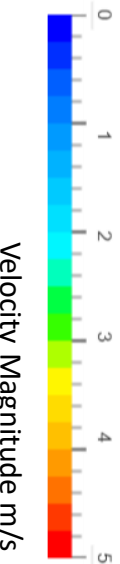
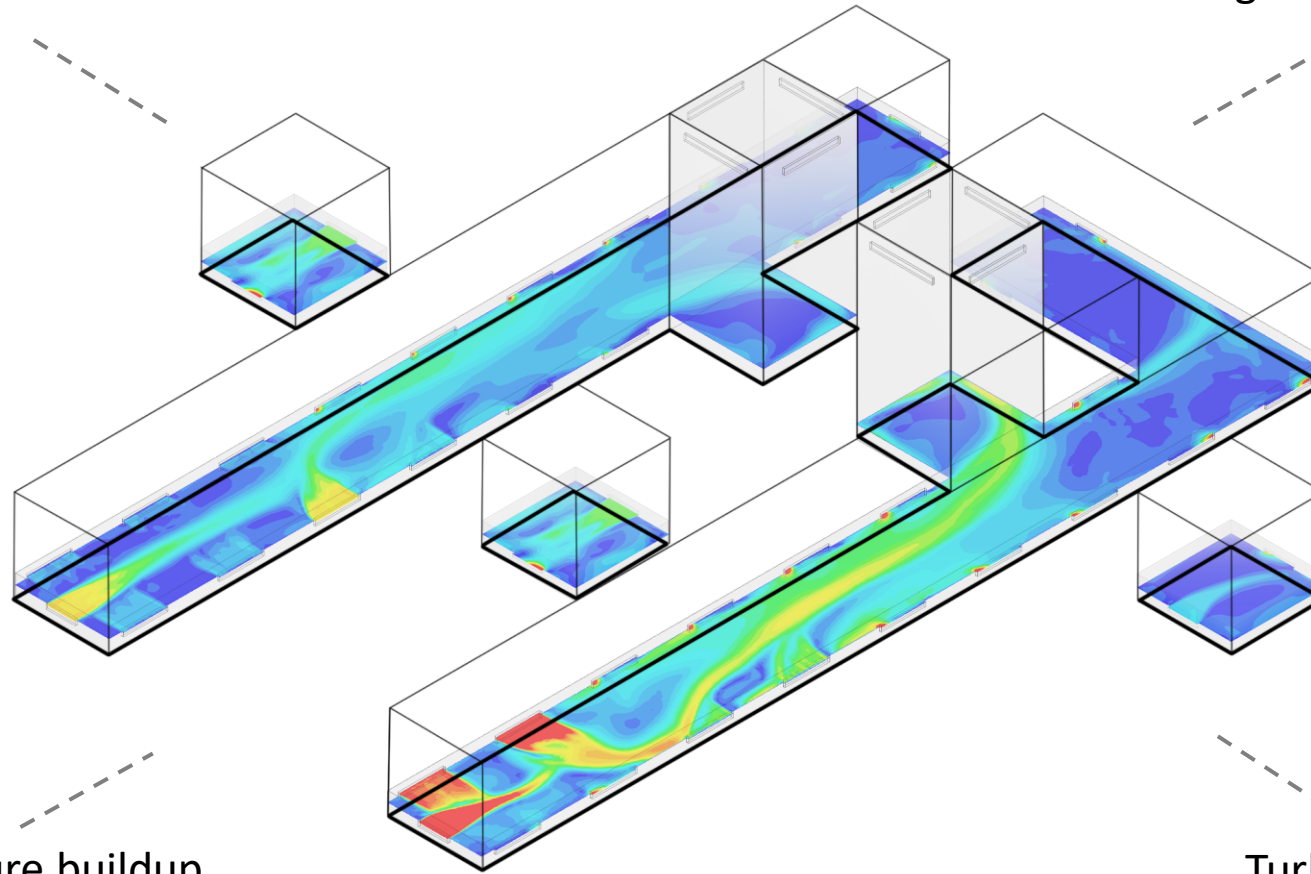
Internal Air Flow Interpretations :

Smooth flow , Strong Cross
Ventilation achieved

Partial Entry, pockets show
stagnation zones

High Pressure buildup
Near wind entry

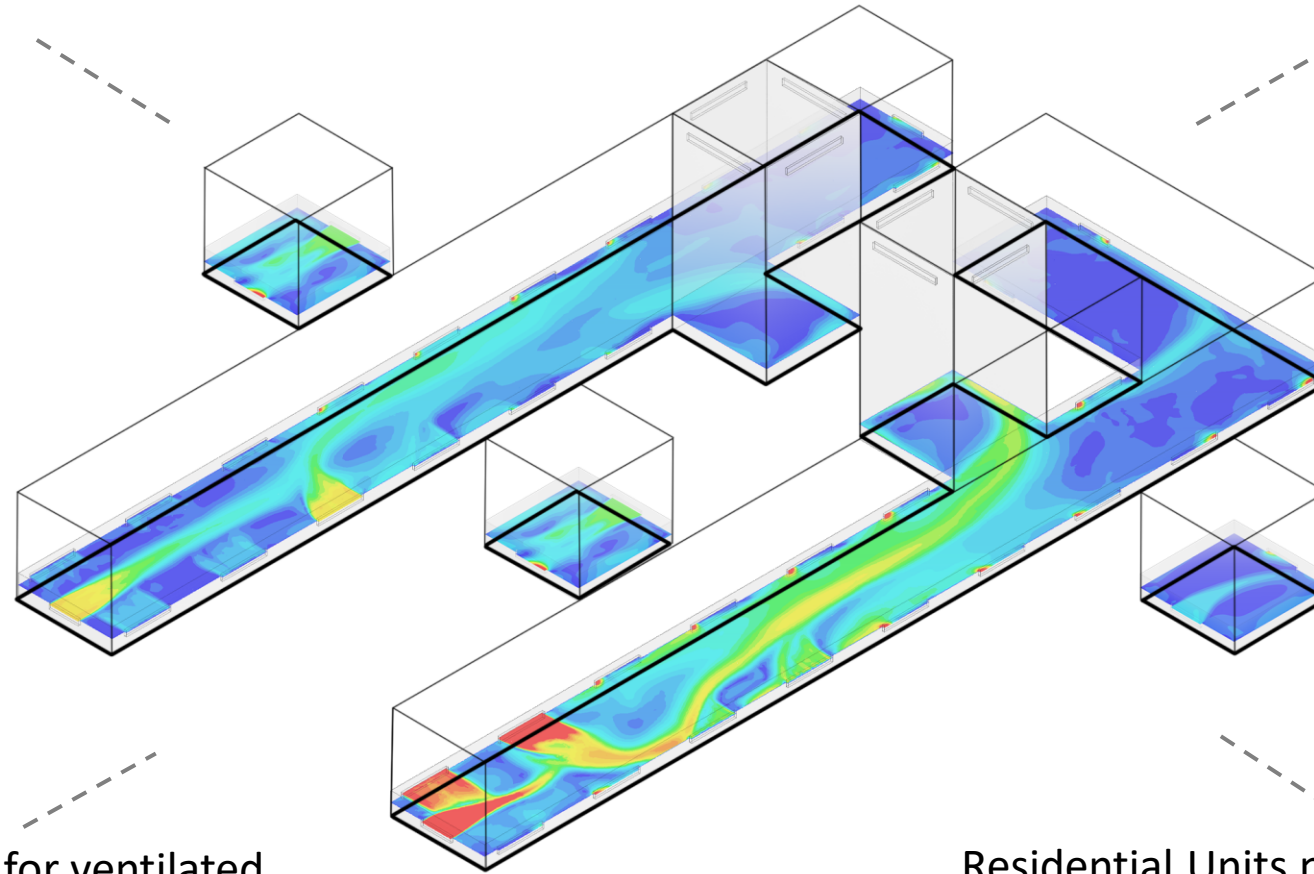
Turbulent flow,
weak internal ventilation



Internal Air Flow Design Decisions :

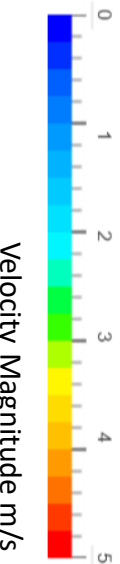
Residential Units planning for optimum cross-ventilation

Semi-open terraces/meeting places



Openings on façade for ventilated corridors

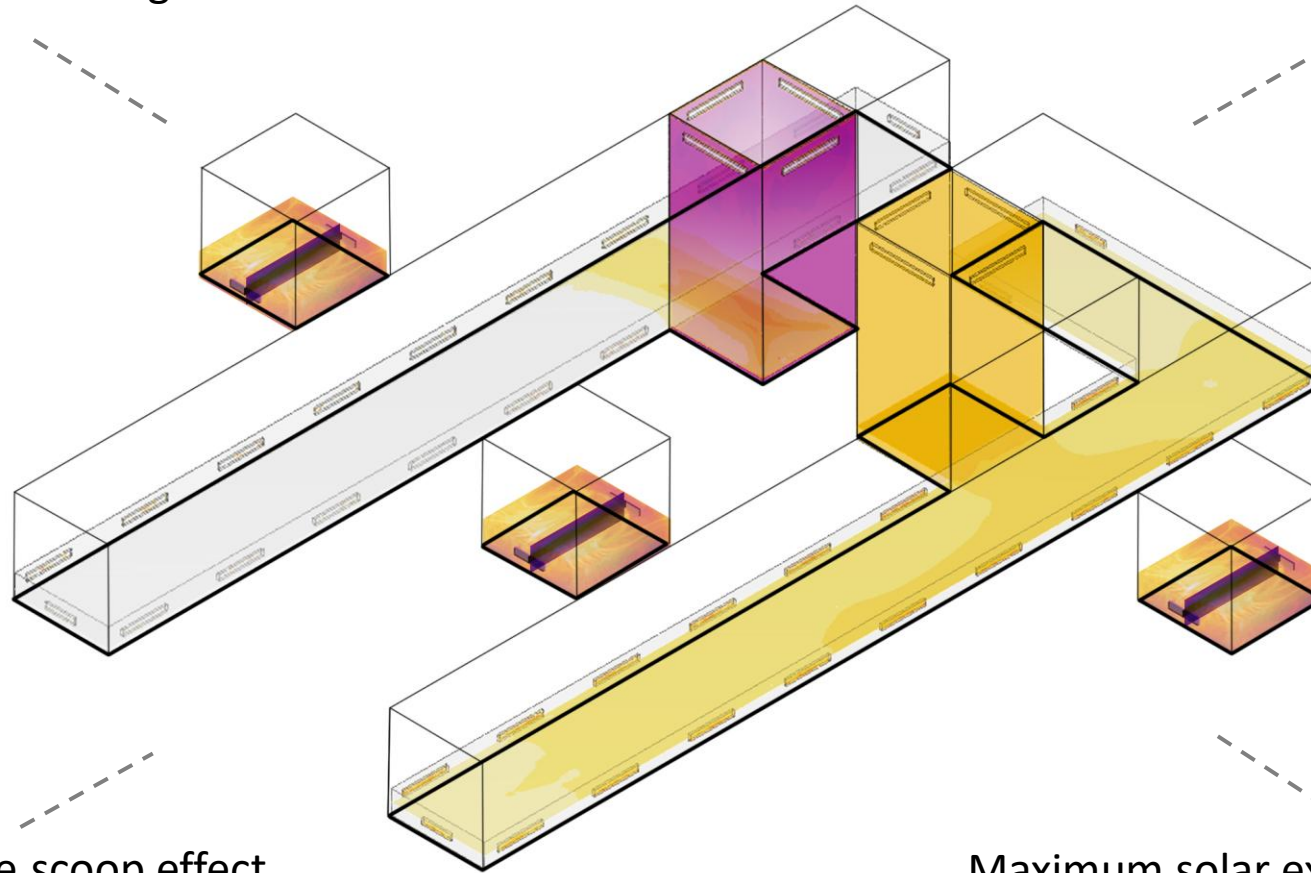
Residential Units planning for optimum cross-ventilation



Internal Heat Flow Interpretations :

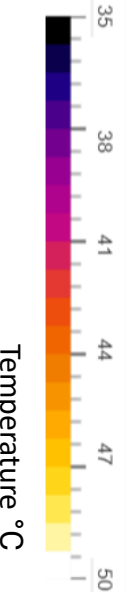
Lower temperature convection due to mutual shading

Wind cools the exo-surface



Façade will experience scoop effect

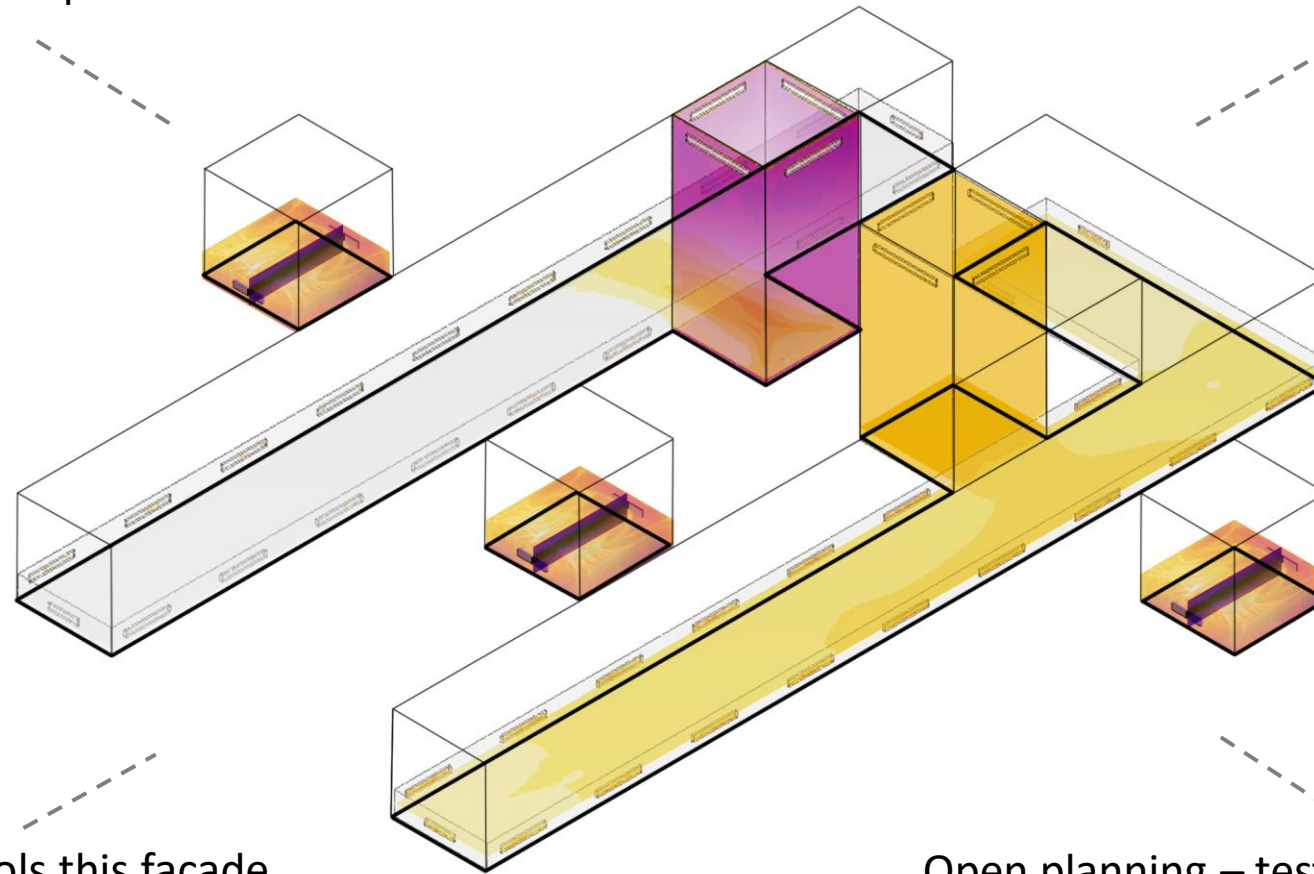
Maximum solar exposure leading to heating of atrium



Internal Heat Flow Design Decisions :

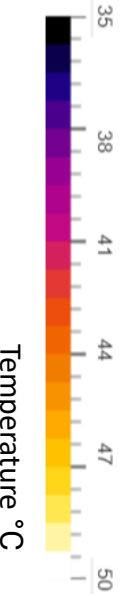
Occupants with enclosed walls i.e. residents shall be placed

Semi-open terraces/meeting places

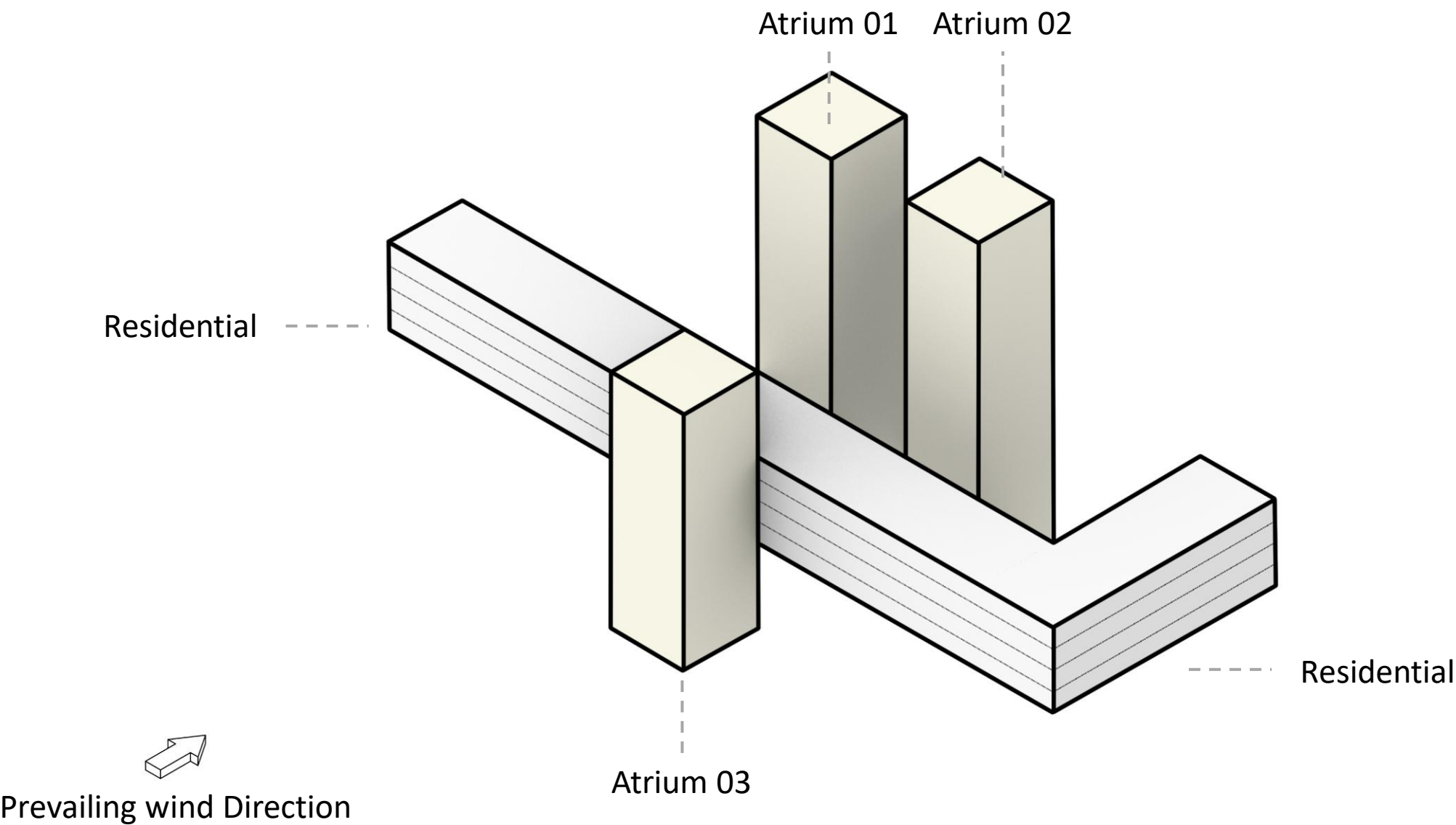


Prevailing wind controls this façade temperature

Open planning – testfit for monk residences



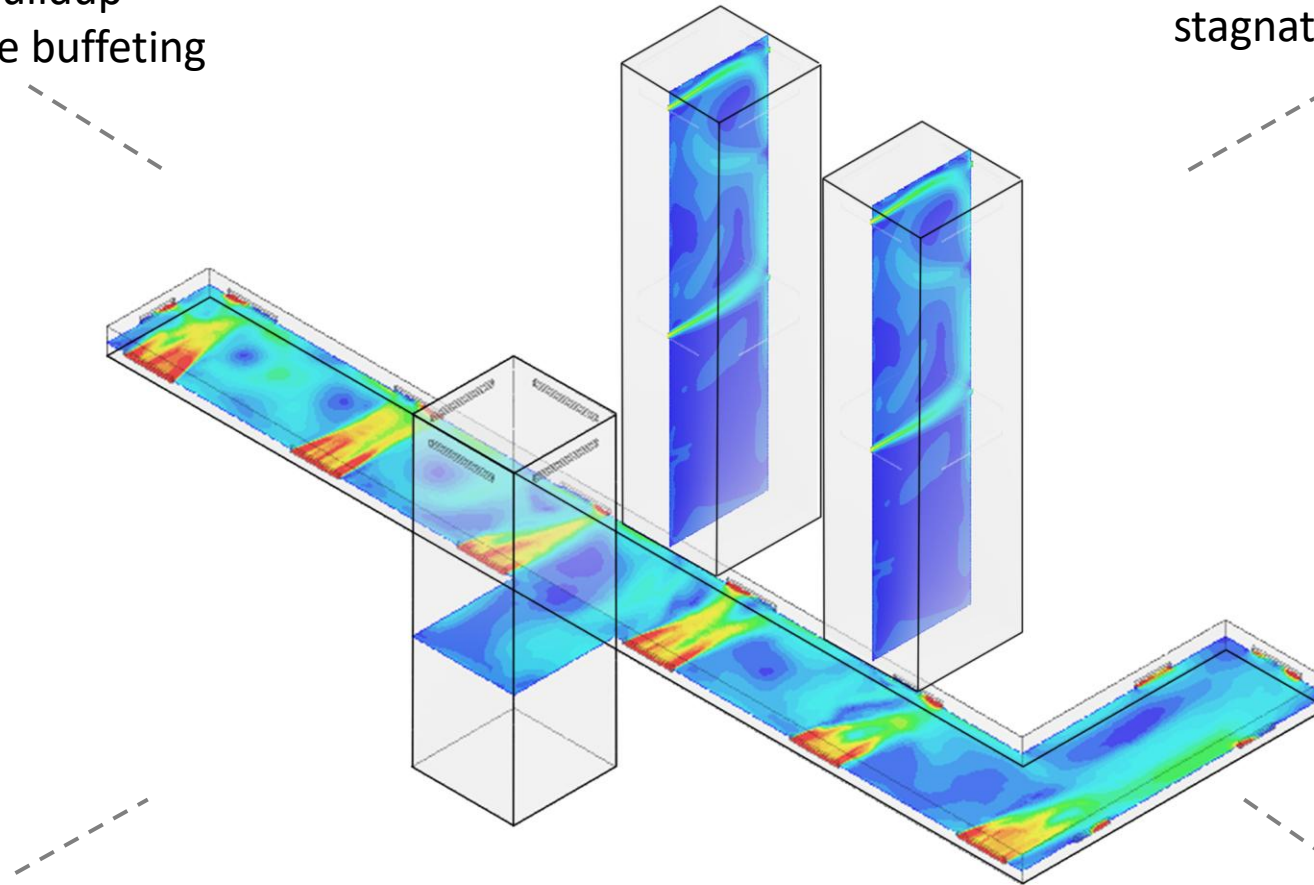
Module for Internal Simulations (Level 73 – 210m) :



Internal Air Flow Interpretations :

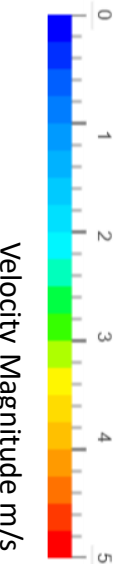
High Pressure buildup
Due to internal wake buffeting

Partial Entry, pockets show
stagnation zones



Smooth flow , Strong Cross
Ventilation achieved

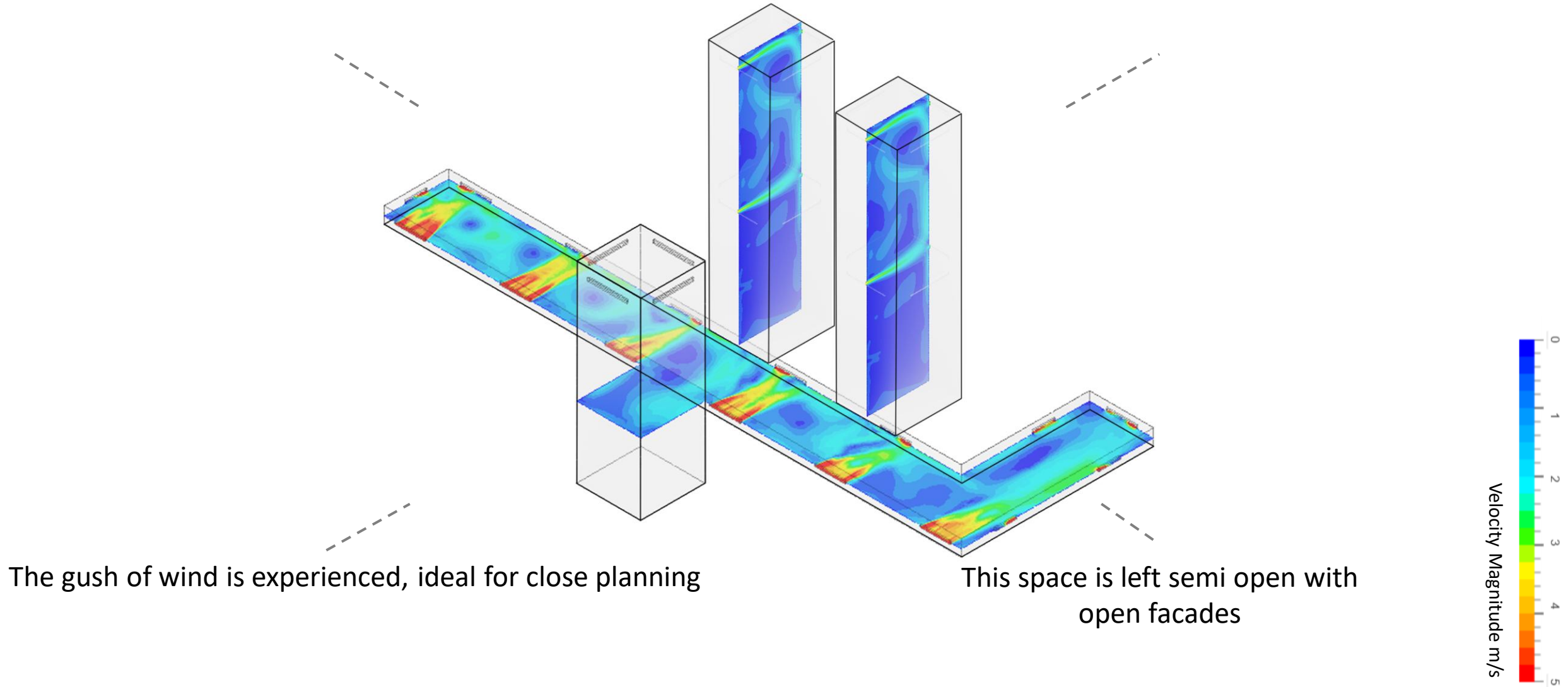
Turbulent flow,
weak internal ventilation



Internal Air Design Decisions :

Internal wake buffeting will increase pressure

Best for open planning for monks



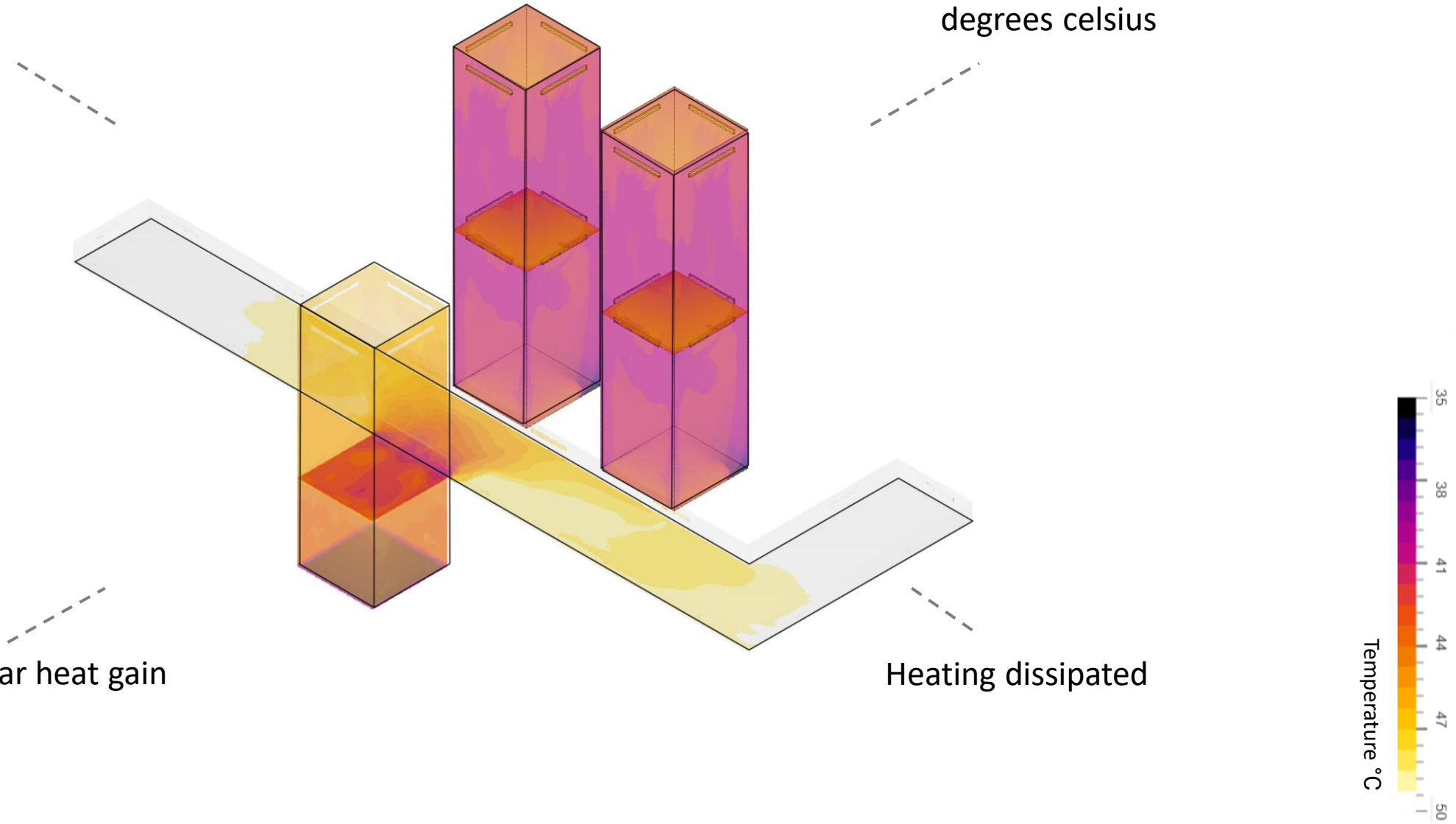
Internal Heat Flow Interpretations :

Mutual Shading from other loops

Atrium from level 71 increases the heating upto 47 degrees celsius

Maximum solar heat gain

Heating dissipated



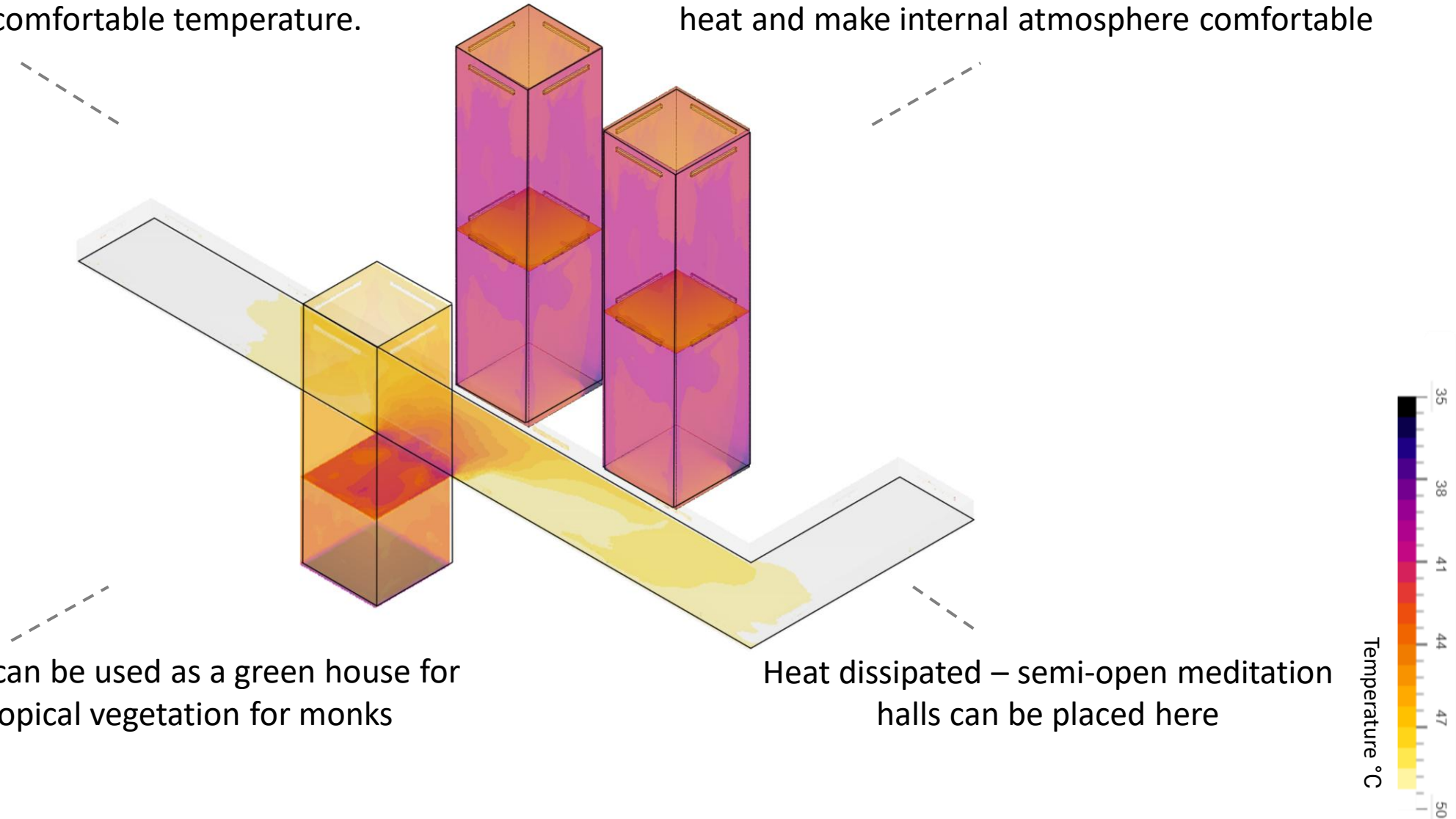
Internal Heat Flow Design Decisions :

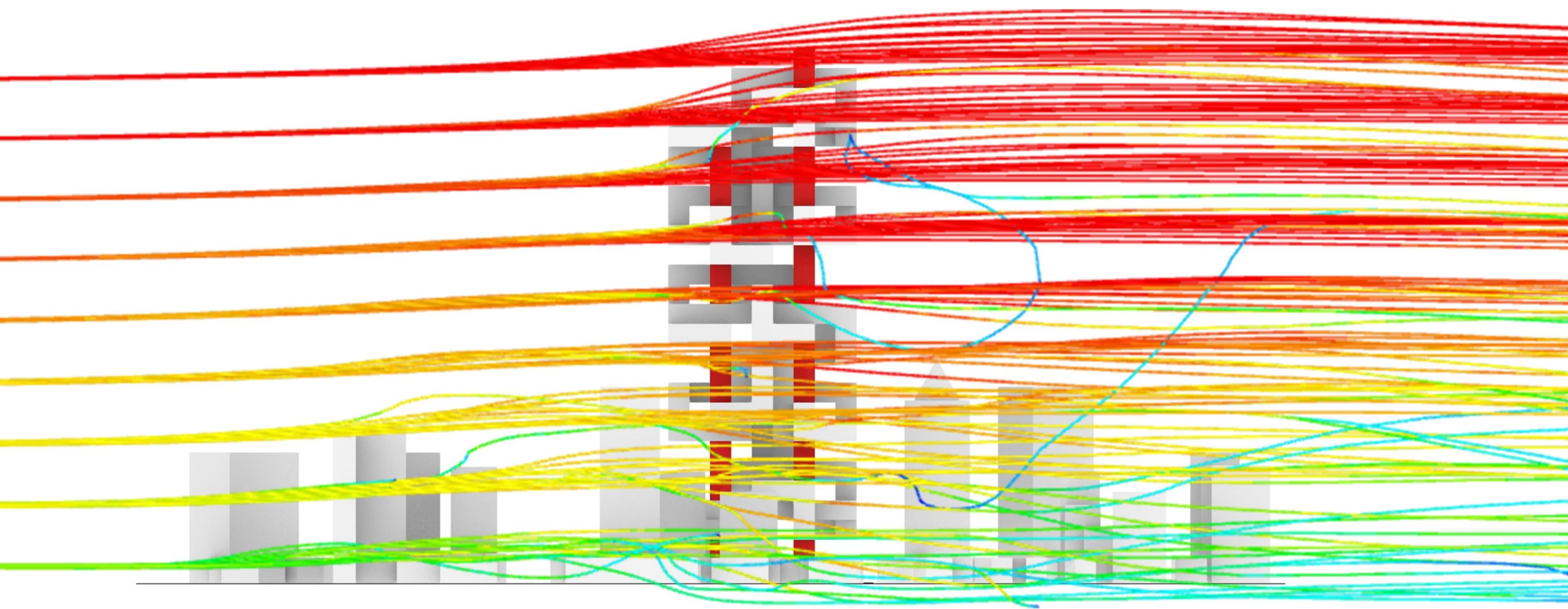
Corridors leading to residences should have opening this side to maintain comfortable temperature.

These atriums need to be ventilated more for neutralizing heat and make internal atmosphere comfortable

Atrium towards this side can be used as a green house for indoor farming for tropical vegetation for monks

Heat dissipated – semi-open meditation halls can be placed here





Thank You